

UNIVERSITY OF SOUTHAMPTON

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SCHOOL OF ECONOMIC, SOCIAL & POLITICAL SCIENCES

DISSERTATION: RESEARCH TOPICS

Economics of Digital Platforms

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I declare that this dissertation is my own work, and that where material is obtained from published or unpublished work, this has been fully acknowledged in the references.

Signed: 31946747 27/04/2023

+ contents

+ appendix 1

+ check sources and finish bibliography

Introduction: explain the case and state clearly the anticompetitive concern/antitrust problem.

Introduction

- Explain the Google-Fitbit acquisition in the context of digital platforms
- State the anticompetitive concern/antitrust problem
- Introduce the concept of digital platforms and their importance in today's economy
- Highlight the potential impact of network effects and economies of scale on competition in digital markets

The European Commission initiated an in-depth investigation into the proposed acquisition of Fitbit by Google, concerned that the deal would further entrench Google's market position in online advertising markets by increasing its access to personal data for ad personalization.

The undertakings concerned have a combined aggregate world-wide turnover of more than EUR 5 000 million (Google: EUR 144 580 million; Fitbit 1 282 million)⁵.⁵ Turnover calculated in accordance with Article 5 of the Merger Regulation. 13 Each of the undertakings has an Union-wide turnover in excess of EUR 250 million (Google: EUR [...] million; Fitbit: [...] million), but they do not achieve more than two-thirds of their aggregate Union-wide turnover within one particular Member State.

Add a breakdown of Google's market share, previous anticompetitive cases, and Fitbit's market share.

Plaintiff: why the acquisition is anticompetitive.

- Economies of scale and scope
 - Network effects/Two-sided markets
 - Gatekeeper and essential facilities
 - Excessive pricing
 - Self--preferencing
 - Exclusionary agreements
-
- Explain the role of network effects in digital platforms and how they can lead to barriers to entry and reduced competition
 - Highlight the potential for Google to leverage its data advantage across markets and reduce innovation and choice for consumers
 - Discuss concerns about vertical integration and privacy for consumers
 - Consider the potential impact of the acquisition on the competitive landscape in the wearable and fitness market

Abuse of dominant position and mergers regulation

Mergers regulation (Issues directly related to Fitbit being operated by such a large player)

Large mergers between companies with greater than €5,000M worldwide turnover and €250M EU turnover need to obtain clearance. While Fitbit did not fall under this, it was still a sizable company with **X many users**.

Introduction

Context

The EC is initiating an in-depth investigation into the proposed acquisition of Fitbit by Google, over concerns that the deal would further entrench Google's online advertising dominance and reduce welfare for customers in the wearables market.

Acquiring Fitbit would provide Google with Fitbit's user health and fitness data and the technology to develop a similar database. Data collected via wearables could be a significant advantage in online advertising markets, making it difficult for rivals to match Google's services. This could raise entry and expansion barriers for competitors, ultimately leading to higher prices and less choice for advertisers and publishers. **The commitment of Google to silo certain wearable data restricted from advertising purposes, (as per the report; cite), is considered insufficient.** The

Commission now has 90 working days to decide, and the opening of an in-depth inquiry does not prejudice the final result of the investigation.

The investigation also examines the effects of combining Fitbit's and Google's databases and capabilities in the digital healthcare sector, which is still nascent in Europe, and whether Google would have the ability and incentive to degrade rivals' wearables' interoperability with Android OS once it owns Fitbit.

- The health monitoring wearables market, worth \$30bn in 2019, is projected to grow at a 26.4% CAGR through 2027 (Fortune, 2019). This matches the Kondratiev (1925) wave growth phase, marked by rapid technological innovation and robust market demand.

Google's acquisition qualifies for EC oversight given that turnover is above **\$5bn and assets are above X**. Fitbit has 1700 employees, and was founded in 2007 (Crunchbase, 2021).

Fitbit's 4.2% market share (Economic Times, 2020), they have actively responded to competition from Apple (34.1%), Xiaomi (12.5%), Huawei (9%), and Samsung (8%).

MENTION 1) INTRA-INDUSTRY IMPACTS; Market concentration and abuse of deep pockets

MENTION 2) EXTRA-INDUSTRY IMPACTS; Advertising Monopoly and Ecosystem

As of 2020, Fitbit has 1,700 employees and holds 4.2% market share in the wearable market (Crunchbase, 2021), while Google represents WearOS, with 3% software share (Counterpoint, 2021).

The EC's investigation also explores the effects of combining Fitbit's and Google's databases and capabilities in the nascent digital healthcare sector in Europe and whether Google would have the ability and incentive to degrade rivals' wearables' interoperability with Android OS once it owns Fitbit. The health monitoring wearables market, valued at \$30bn in 2019, is projected to grow at a 26.4% CAGR through 2027.

EVALUATION OF ARGUMENTS. MAXIMISATION OF WELFARE JUDGEMENT.

REMEDIES PROPOSED.

While Google may improve their offering by acquiring Fitbit, it is valid that they can start a competitor as an extension to the Pixel range. This would increase competition in the market, and lead to dynamism.

Google's history of self-preferencing is bad.

Google's WearOS is advantageous and may result in an incumbent joining the software...

Apple's offering is technically superior and there are cheaper substitutes that offer interoperability...

Judge: Propose remedies/fines, considering the implication that such remedies/fines may have on the market and consumers.

- Weigh the arguments from both the plaintiff and the defendant, considering the impact of network effects and economies of scale on competition in digital markets

- Discuss potential remedies or fines, considering their impact on the market and consumers
- Consider the potential consequences of allowing or blocking the acquisition on competition and innovation in the wearable and fitness market
- Highlight the importance of balancing the benefits of digital platforms with the need for competition and consumer protection

Data separation: Google committed to not using the health and wellness data collected from Fitbit devices for Google Ads. This would help ensure that the data advantage concern was addressed.

Data access for third parties: Google agreed to maintain access to Fitbit users' data for third-party software and services, ensuring that other companies in the market can continue to develop and provide services that integrate with Fitbit devices.

Interoperability: Google committed to maintaining the interoperability between Fitbit devices and third-party wearables and devices that use Google's Android ecosystem, ensuring that other competitors can still access the platform and compete effectively.

Previous concerns like Google's acquisition of DoubleClick in 2007 was met with significant scrutiny from both the FTC in the U.S. and the European Commission. Ultimately, both regulatory bodies approved the deal, but the European Commission's approval came with conditions to protect competition in the online advertising market. They have only served to drop the cost of CPM.

- In 2011, Google acquired ITA Software, a company that provided software for airline reservations and flight data. The DOJ approved the acquisition but imposed conditions to ensure competition was maintained in the online travel search market.
1. Google's acquisition of AdMob in 2010 faced scrutiny from the FTC, which was concerned about the potential anticompetitive effects in the mobile advertising market. The acquisition was eventually approved after the FTC concluded that other competitors in the market were emerging, and the deal would not significantly harm competition.

Introduction

This study aims to evaluate whether the Google's acquisition of Fitbit will maintain long-term competition and consumer welfare in wearables and online advertising markets. This study aims to take an economic perspective on intra-industry impacts, such as market concentration, abuse of dominant position, and extra-industry impacts, including Google's advertising strength and software ecosystem, to assess whether the acquisition should take place, and if so, under what conditions.

The acquisition qualifies for European Commission (EC) oversight as combined turnover exceeds €5,000M, with Google's turnover at €144,580M (Alphabet, 2019) and Fitbit at €1,282M (EC, 2021), with EU-wide turnovers both exceeding €250M. The EC enact antitrust policy to favour consumers by encouraging companies to offer goods and services in the most favourable terms (EC, 2021). Opposition from the Competition and Marketing Authority arose over concerns that the deal would reduce competition and further consolidate Google's market dominance in online advertising, with better access to health data for ad personalisation (Balogun & Wadlow, 2020; BEUC, 2020).

The competition authority, the plaintiff, reminds the EC that Google has previously received anticompetitive fines over self-preferencing, and argues that the acquisition is anticompetitive due to vertical integration and economies of scale, market concentration, and the added potential for excessive pricing, economies of scale, and exclusionary agreements. Google, the defendant, argues that the acquisition will improve product offering, lower barriers to entry through WearOS, and help improve service experience for millions of users while ensuring consumer trust and data security. These arguments are considered in relation to their short- and long-term effects on the market.

Argument of the Plaintiff

Argument 1: *Fitbit independence is essential for maintaining competitive constraints on wearables.*

1.1. Market concentration and Threat of Response

Firms venturing into the wearables industry experience substantial barriers to entry; enticing users to switch from well-established platforms demands a strong platform with developer support and audience, meaning entrenched networking effects must be overcome to generate sales. Of existing operating systems entrants could otherwise adopt, Google's WearOS offers great interoperability capabilities with Android (Srinivas, 2021; IDC, 2019), leading rivals and new entrants like Fossil, Mobvoi, and Misfit to adopt the software and encourage ecosystem development and wider audience reach (WearOS, 2020).

Fitbit overcame these barriers to entry as an industry pioneer with Fitbit OS, strategic acquisitions and an aggressive product launch schedule (including smartwatches since 2017) (IDC, 2019), increasing their bargaining power over suppliers and strengthening brand recognition (BEA.gov, 2021). They have exhibited high dynamic-x by multi-sourcing 70% of outsourced parts (Delmas, 2018) and working with many separate manufacturing sites for assembly, supporting an agile supply chain, as per Porter's Five Forces (1980).

If Google acquires Fitbit, all remaining major players will be resource-rich corporations, setting the stage for big tech to seize the lion's share of growth as the market matures. Market concentration, measured by the 6-firm HHI, would shift from 1,490.7 to 1,506.5 (Appendix 1). This change signifies a deterioration from a highly competitive environment to a moderately competitive one, according to the standard threshold of 1,500 set by the DOJ (2018). Incumbents can raise barriers to entry by bolstering their competitive advantage through networking effects, economies of scale, tacit collusion, and threat of incumbent response. Further knock-on effects for market dynamism may manifest in price-fixing and predatory pricing (HBR, 2019; Dayen, 2020).

1.2. Self-preferencing

With Google controlling both the hardware (Fitbit devices) and software (WearOS), greater incentive will form to limit access to the WearOS ecosystem for competition through exclusionary dealings, something they were recently found guilty of in France in 2019 (Reuters). Greater vertical integration means larger suppliers could face pressure to

exclusively partner with major competitors like Google (Belleflamme & Peitz, 2022), preventing entrants from gaining a foothold in the market (McKinsey, 1993).

Improving interoperability between Android and Fitbit watches will increase demand for Fitbit's existing iPhone users to switch to Google's software ecosystem. Since 2010, EC investigations have fined Google €8.25 billion for violations including: improving the ranking of its subsidiaries in Search; adding restrictive clauses in contracts with third-party websites; and requiring smartphone makers to preinstall Google Search and Chrome in exchange for Play Store (EC, 2017; CNN, 2022).

Antitrust policies favour consumers by encouraging companies to offer goods and services in the most favourable terms (EC, 2021). Wearables producers should cater to diverse consumer preferences by offering as wide array of products as possible, and Google is maintaining variety when they could launch a competitor instead, potentially to combat low consumer trust problems (BIS, 2018). In concentrated markets, incumbents encounter fewer obstacles when seeking profit-maximisation, due to tacit collusion, misaligned focus from achieving dynamic-x through R&D investments, innovative product launches, and employee skill enhancement (Bates, 2019).

Argument 2: Google's acquisition exacerbates advertising ecosystem dominance, stifling innovation across markets.

2.1. Strengthened Dominant Position

Google potentially utilises WearOS data, including GPS, sleep duration, and app usage, for targeted advertising and user analysis, creating rich profiles that enable personalized ads and enhance user engagement (Google, 2021). By acquiring direct competitors such as Doubleclick and AdMob, they have strengthened their advantage in search advertising CPM (Cost per Mille), resulting in 'bottleneck market power' for Google and supra-competitive prices (Dinielli, 2021; NYTimes, 2019). Advertising, primarily within services like Android, Chrome, hardware, Maps, Play, Search, and YouTube, accounts for 83.2% (\$134.8bn) of Google's revenue (Alphabet, 2019).

Given Google can develop an Apple Watch alternative, its interest in Fitbit conceivably comes from access valuable databases containing unique health-related data points. Such data can be fed into behavioural detection models to improve engagement for YouTube or Google Assistant (Google, 2021), increasing competitive advantage due to economies of scale and networking effects, and raising barriers to entry. With low brand trust, Google's ownership may reduce consumer welfare and result in boycotting (Observer, 2019).

2.2. Long-Run Impacts of Digital Ecosystem

Currently, Google and Apple hold 99% global duopoly in the smartphone OS market (Taylor, 2020). Both parties have historically limited interoperability between their software, such as iMessage, FaceTime and the Google Play Store, which have erecting barriers for users to switch between platforms and for competition to enter the market (USHOR, 2020). Their shared duopolies extend to web browsers, cloud storage, and media streaming.

Long-term, Google's deep financial resources could be used to implement predatory pricing strategies, forcing exits of limit-resource competitors. The legal battle between Apple and Epic Games concerns "anticompetitive" app store commissions identical to Google Play's (Android, 2020), underscoring the potential for market failure (BBC, 2021).

Antitrust policymakers should be vigilant regarding the consequences of strong network effects; increased dominance by established platforms can lead to greater network value and cheaper prices short-term, but raise barriers to entry for innovative newcomers long-term, harming welfare.

To illustrate this, assume: Benefits of using Google: $b + kx_G$; where b represents benefit of Google's network effects. Benefits of using a new platform (N): $(1-b) + kx_N$, with low network effects but high intrinsic value.

The fraction of users choosing platform N is given by: $1/2 + k/2 * (x_N - 1)$ (see Appendix 2). Under strong network effects, where $k \gg 1/2$, Google's ability to attract users becomes more significant with greater connections and interactions, developer incentives, and reduced marginal costs, while intrinsic benefits of competitive platforms become less important.

Entrenched loyalty and switching costs further create market failure by reducing ability to gain traction for entrants. Consequently, consumers may face reduced choice and default to dominant players, producing deadweight loss (HBS, 2019; Ofcom, 2020). Ultimately, market velocity creates long-term benefits that arguably supersede the value of reduced platform costs for consumers, and policy-makers should protect competition with this ruling.

3.0. Argument of the Defendant

Google's acquisition is highly beneficial to consumers

Google's acquisition of Fitbit creates positive externalities that are three-fold: enhanced competition between wearables, lower barriers to entry through WearOS, and greater third-party benefits across Google services.

First, Google can heighten competition by bringing complimentary expertise to the Fitbit product line. In the smartwatch industry Apple holds a substantial dominant position, characterised by robust brand loyalty, an extensive ecosystem, and feature-dense products such as cellularly-connected smartwatches (Koetsier, 2021; EC, 2021). Consumers may opt for Apple Watch products, despite the potential suitability of alternative options that better align with individual needs or preferences, reflected in pricing strategy that is markedly higher than comparable alternatives (Mundy, 2022; IDC, 2019).

Google's mobile software control, artificial intelligence expertise, and data analysis capabilities will be used to heighten Fitbit product strength, with cellular-connectivity, clinical alerts, and competitive prices, benefiting consumer surplus (Forrester, 2019). With greater economies of scale due to Google's flagship smartphone Pixel 5, greater elimination of redundancies and costs for consumers becomes possible (Google, 2020b). As with Waze (Empson, 2013), Google can improve product performance and market variety, internalising dead weight loss and competing away monopoly profits.

In support of this, Google has consistently produced services more cost-efficiently than competitors like Apple (Bitton & Lewis, 2020; NYTimes, 2020), as evidenced by the lower price points of Android smartphones and the costless provision of many of its digital platforms (Wired, 2023). In contrast, Apple has utilized its market advantage to intentionally slow-down device performance (BBC, 2020), potentially seeking repeat purchases and profit maximisation.

Second, Google is uniquely positioned to provide a robust alternative to Apple while enhancing competition across markets and reducing barriers to entry. While Apple has designed its systems to be generally interoperable, Google makes no products wholly exclusive to Android (Android, 2020; IGN, 2022).

By acquiring Fitbit, Google can align compatibility between hardware and software for the WearOS system at large, addressing the negative perceptions that have been previously associated with the platform (Bohn, 2021). This acquisition may increase the value in investing in WearOS, subsequently boosting the platform's intrinsic value. A more powerful platform will better enable new market entrants to compete with established original equipment manufacturers, such as Apple's watchOS, Samsung's TizenOS, and Huawei's HarmonyOS.

Third, Google will be able to refine its offerings for millions of users globally by leveraging the wealth of data generated by wearables. By leveraging this data, Google can algorithms can further be optimised for consumer satisfaction within Google Search, Maps, Android, Play, and YouTube, as well as associated APIs, which are used by well over a billion people (Insider, 2015), and in-line with Fitbit's data-sharing for multi-homing via API (Fitbit, 2019; EC, 2021).

Fitbit's recent data scandals (Fowler, 2021) may have reduced consumer trust, and better protection is also offered by Google's sophisticated security infrastructure. Google can create secure silos (NYTimes, 2019b), where sensitive data like profile enrichment are not auctioned on AdWords, while staying protected, improving service quality for users.

Judgement

In the interest of making the wearables market as efficient as possible, the court will attempt to strike balance between competing interests. Recognising the strength of incumbents, there is a credible argument suggesting a meticulously-structured acquisition could serve to internalise dead weight loss and maximise welfare. With necessary commitments from Google, the acquisition may proceed to bring competition to wearables in the long-run.

Firstly, Google's promise to maintain open data access for third-parties, by allowing other companies to develop on the WearOS platform and offering integration with Fitbit devices through consensual access to share data with applications, the wearables market can increase diversity and competition, promoting product variety and preventing market monopolization by Apple or Google.

Secondly, sustained interoperability and device compatibility remedies potential monopolistic behaviour, fostering competition and consumer welfare. By maintaining smoothness between Fitbit devices and smartphones using Google's Android ecosystem, self-preferencing and exclusionary practices are limited, promoting market diversity.

Lastly, Google must commit to data separation and develop silos that prevent sensitive health data from Fitbit from being used to unfairly strengthen Google's advertising business. This directly addresses concerns raised by the plaintiff, preventing the direct use of Fitbit data to alleviate fears of market failure. This aims to maintain and strengthen consumer trust, which is essential for a efficacious market.

Appendix 1:

- State the market shares used and the calculation conducted. 6th firm was 2.5% I believe.

Appendix 2

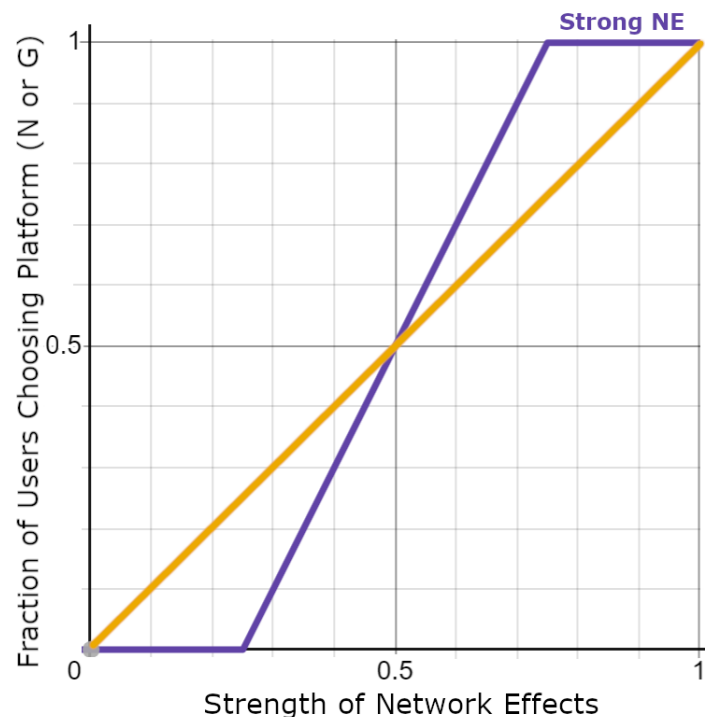


Figure 1.

Strength of Network Effects against Fraction of Users Choosing Platform (N being new entrants, or G, being Google's ecosystem. Strong networking effects are illustrated with the purple curve.

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